7. ORBITGUARD - Decentralized Space Debris Tracking

ORBITGUARD seeks to democratize space safety by using small satellites equipped with optical telescopes to track space debris. By moving away from slow, government-monopolized tracking, this open-access database allows satellite operators to query collision risks and coordinate avoidance maneuvers autonomously via smart contracts.

Mission Brief

Design a constellation of small satellites with optical telescopes that track space debris and create a decentralized, open-access database where satellite operators query collision risks and coordinate avoidance maneuvers via smart contracts. Current debris tracking is monopolized by governments (slow, restricted access).

Satellite Requirements

- **Orbit:** Multiple planes (LEO 400km, 600km, 800km) to observe debris from different altitudes
- **Payload:** Optical telescope (20cm aperture, tracks objects down to 10cm in LEO)
- **Coverage:** Continuous monitoring of high-traffic orbital shells
- **Power:** 40W average (telescope + star tracker + onboard processing)
- **Mass:** <50 kg per satellite (12U CubeSat with deployable optics)
- **Comms:** S-band for telemetry + debris catalog updates

Web3 System

Data Flow: Satellite optically tracks debris → Calculates TLE (two-line element) orbit data → Submits to decentralized debris catalog (stored on-chain + IPFS) → Multiple satellites confirm same debris object (consensus) → Satellite operators query collision probability for their spacecraft → If Pc > 0.0001, smart contract suggests optimal avoidance maneuver → Operators vote on maneuver coordination (if multiple sats affected) → Blockchain records all maneuvers for accountability

Key Smart Contracts:

1. **DebrisRegistry** - Decentralized catalog of all tracked objects with TLE data
2. **CollisionPrediction** - Calculates conjunction events, proposes maneuvers
3. **ManeuverCoordination** - Multi-satellite voting for coordinated avoidance
4. **DataContribution** - Rewards satellites for accurate debris tracking
5. **OperatorReputation** - Tracks which operators perform maneuvers vs ignore warning

Round 1 constraints:

1. Mission Objectives & Success Criteria

Every design must start with the **"Why."** In a hackathon, your mission objective isn't just a goal; it's a boundary.

- **Operational Lifetime:** How long must the satellite survive? (e.g., 6 months for a tech demo vs. 5 years for a weather satellite).

- **Target Orbit:** This dictates almost everything else.
    - **LEO (Low Earth Orbit):** Lower latency, higher resolution, but high atmospheric drag.
    - **GEO (Geostationary Orbit):** Constant coverage of one area, but extremely high launch costs and radiation.
- **Data Latency:** Does the data need to be "real-time" (for disaster relief) or can it be stored and forwarded once a day?

2. Payload Constraints

The payload is the "customer" of the satellite; everything else (power, thermal, comms) exists just to keep it happy.

- **SWaP-C:** This is the golden rule of satellite design: **S**ize, **W**eight, **a**nd **P**ower - **C**ost.
- **Mass & Volume:** Most hackathons use the "U" standard (1U = 10x10x10 cm). You'll likely be restricted to a 3U or 6U form factor.
- **Duty Cycle:** Can the payload run 24/7, or does it need to shut down to recharge batteries while the satellite is in the Earth's shadow?

3. Budget & Resource Estimation

In a real-world scenario (and a good hackathon), "unlimited money" isn't a thing.

- **Total Cost of Ownership (TCO):** You must account for:
    1. **Development:** Components and labor.
    2. **Launch:** Often calculated per kilogram (e.g., $50,000/kg for LEO).
    3. **Operations:** Ground station fees to talk to your satellite.
- **COTS vs. Custom:** Using **Commercial Off-The-Shelf (COTS)** parts is cheaper and faster but might not be "space-hardened" against radiation compared to custom, radiation-hardened components.

4. Technical Constraints & Requirements

1. Decentralized Surveillance: The Optical Network

- **Distributed Observation:** Unlike ground-based radar that struggles with "atmospheric noise," ORBITGUARD uses a constellation across multiple orbital planes (**400km, 600km, 800km**). This provides a "multi-angle" view of high-traffic LEO shells, allowing for more precise tracking of small objects.
- **Payload Precision:** Each 12U CubeSat carries a **20cm aperture optical telescope**. This hardware is specifically designed to track "sub-catalog" debris—objects as small as **10cm**—which current government systems often miss or report with high latency.
- **Onboard Intelligence:** The satellites utilize **Edge-AI** to calculate **TLE (Two-Line Element)** data in real-time. By processing data in orbit, the system only downlinks critical orbital updates, significantly reducing the "detection-to-alert" window.

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2. The "DebrisRegistry": An Immutable Catalog

- **Trustless Consensus:** To ensure data accuracy, ORBITGUARD employs a **Multi-Satellite Consensus** mechanism. A debris object is only added to the official on-chain registry once its trajectory is confirmed by at least three independent satellites.
- **Decentralized Database (IPFS):** High-resolution tracking logs and TLE history are stored on **IPFS**, with their cryptographic hashes recorded on the blockchain. This creates an **Open-Access Catalog** that no single government can restrict or censor.
- **Transparency:** Every orbital update is timestamped, ensuring that satellite operators have a verifiable "Chain of Custody" for debris data when making high-stakes safety decisions.

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3. Autonomous Coordination via Smart Contracts

- **Collision Prediction Oracle:** The **CollisionPrediction** contract acts as a neutral "referee." When two satellites are on a collision course (Probability of Collision $P_c > 0.0001$), the contract automatically calculates the most fuel-efficient avoidance maneuver for both parties.
- **Democratic Maneuvering:** Through the **ManeuverCoordination** contract, operators of affected satellites can vote on coordinated plans. This prevents "chicken" scenarios where both satellites move in the same direction, inadvertently maintaining the collision risk.
- **Reputation & Accountability:** The **OperatorReputation** contract records who follows safety warnings. This creates a public "Safety Score," incentivizing responsible orbital behavior and potentially lowering insurance premiums for high-rated operators.

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4. Engineering & Deployment Viability

- **Scalable 12U Platform:** Using a <50 kg mass profile ensures low launch costs. The "deployable optics" design allows a large-aperture telescope to fit within a compact CubeSat frame during launch.
- **Power Efficiency:** The system is optimized for a **40W average**, balancing the needs of the star-trackers (for orientation), the main telescope, and the onboard processing units.
- **Operational Incentives:** Through the **DataContribution** contract, private satellite owners who share their own tracking data with the network earn tokens, creating a self-sustaining and growing database.